EXPERIMENT 5

Design and configure the Switch network using cisco packet tracer

AIM :

To develop a Python program that generates Hamming Code for a given binary data stream.to insert check bits at appropriate positions and ensure even parity for reliable data transmission.

PROCEDURE :

1. Open a Python programming environment and input the binary data.
2. Calculate the number of check bits using (2^r \ge m + r + 1).
3. Insert check bits at positions **1, 2, 4, 8, ...** (powers of 2).
4. Fill the remaining positions with the data bits.
5. Calculate each check bit using **even parity**.
6. Combine data bits and check bits to generate the **final encoded Hamming bitstream**.

CODE:

HAMM.py - C:/Users/Tcs/AppData/Local/Programs/Python/Python310/HAMM.py (3.10.6)

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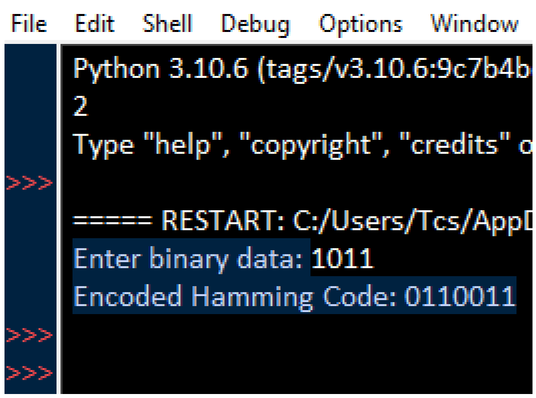
```python
def calculate_redundant_bits(m):
    r = 0
    while (2 ** r) < (m + r + 1):
        r += 1
    return r
def hamming_encode(data):
    m = len(data)
    r = calculate_redundant_bits(m)
    total_length = m + r
    encoded = ['0'] * (total_length + 1)
    j = 0
    k = 0
    for i in range(1, total_length + 1):
        if i == 2 ** j:
            encoded[i] = '0'
            j += 1
        else:
            encoded[i] = data[k]
            k += 1
    for i in range(r):
        pos = 2 ** i
        count = 0
        for j in range(1, total_length + 1):
            if j & pos and encoded[j] == '1':
                count += 1
        if count % 2 != 0:
            encoded[pos] = '1'
    return ''.join(encoded[1:])

data = input("Enter binary data: ")
encoded_data = hamming_encode(data)

print("Encoded Hamming Code:", encoded_data)
```

OUTPUT :



RESULT :

The Python program for the Hamming Code Sender Utility was successfully executed, and the encoded bitstream with even parity was generated.